

Introduction

- ✓ Subarea of cryptology. Cryptology is the scientific area dedicated to the protection of information through the conversion of data into secret and secure formats.
- ✓ Cryptography that studies and implements mechanisms for secure communication between two entities, admitting that communication can be intercepted.
- ✓ Communication involves two phases:
 - Encryption. Transforming the message into something unintelligible (encrypted message)
 - Decryption. Transforms the encrypted message into an intelligible message
- ✓ The passage between the two phases is only possible using the cipher (mathematical algorithm). This algorithm uses a key only known to the participants. Only knowledge of the algorithm and the key allows decryption.

History

One of the first ciphers was the one used by Julius Caesar for communication between the legions. The cipher consisted of advancing x places in the alphabet and the key was three places.

During the Middle Ages, Muslim civilization devoted itself to a serious study of cryptography

From the Renaissance to the modern age, Western civilization began to dedicate itself to cryptography, usually with applications in communications during wars.

Fundamentals of Cryptography

This development culminated in the Germans' construction of the enigma machine. The algorithm was provided by the construction of the machine, but the key was given by the position of rotors that changed daily.

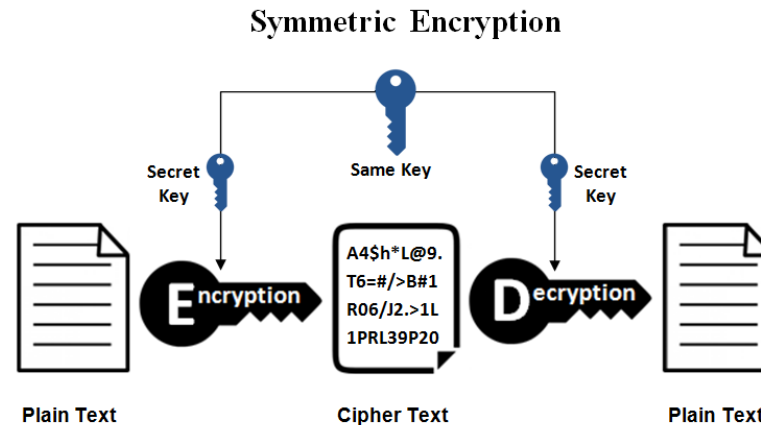


The enigma marked the beginning of *modern hacking* with the construction of the Colossus that was modeled after the Turing machine, i.e., it was a computer.

Symmetric encryption

- ✓ It corresponds to the "classical" view of cryptography. The sender and receiver hold the same key that encrypts and decrypts. As only they know the key, it is called private key or *secret key*.

Fundamentals of Cryptography



<https://www.ssl2buy.com/wiki/symmetric-vs-asymmetric-encryption-what-are-differences>

- ✓ The most widely used symmetric encryption algorithm is AES.
 - Used as standard in the USA since 2002!!
 - There are a few variants: AES-128, AES-192, AES-256 according to the number of bits of the private key.
 - The algorithm puts the information in a matrix and performs transformations: It replaces the information with another, mixes the rows, mixes the columns;

Fundamentals of Cryptography

- 15 algorithms that perform the operations described above were proposed and in the end the Rijndael algorithm was chosen.

✓ Examples:

- Messaging Software: WhatsApp
- Disk encryption: Bitlocker

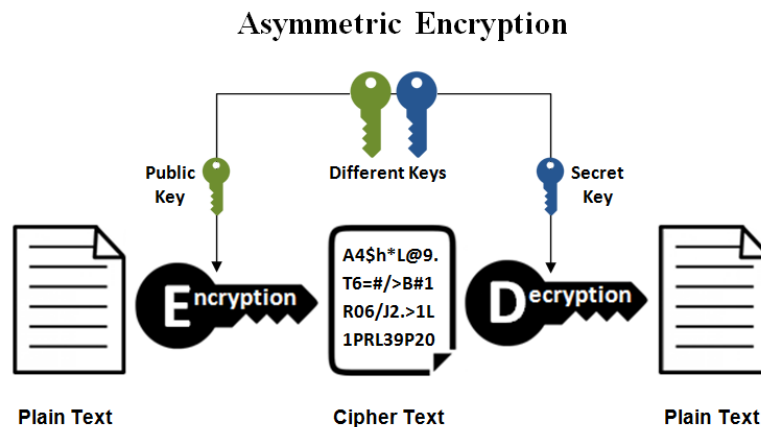
✓ Advantages:

- Simpler = > more efficient

Asymmetric encryption

- ✓ The sender holds a public key (any user can have access to it) and encrypts the message. The receiver holds a private key and can decrypt the message.
- ✓ The person who signs the message uses a private key, the person who verifies the message uses the public key to confirm the message.

Fundamentals of Cryptography



<https://www.ssl2buy.com/wiki/symmetric-vs-asymmetric-encryption-what-are-differences>

Fundamentals of Cryptography

✓ Algorithms

- RSA (Rivest–Shamir–Adleman) x – Currently in version 3, it is the most widely used algorithm for key generation. It presents a very complex key derived from mathematical expressions with prime numbers, where x is the size of the key (at least 2048 bits). This algorithm is used for: digital signatures and message encryption. There are more algorithms such as: DSA, ECDSA, Ed25519, but less used.
- Elliptic Curve Cryptography – Considered a possible successor to RSA. The algorithm is based on the solution of the equation $x = g^y \pmod{p}$ (what is y knowing x , where g and p are secret scalars, such that x and g^y divided by p have the same remainder. This algorithm allows for much smaller keys and is more efficient. However, it was called into question when the backdoor in the algorithm proposed to NIST by the NSA became public.

Fundamentals of Cryptography

Features	RSA	ECC
Security-based	Factoring of large composite numbers	Discrete logarithm problem of elliptic curves
Key size (in bits)	Highest for equivalent security 3072 RSA = 256 ECC	Smaller for equivalent safety
Performance	May be slower for comparable security	Generally faster for comparable safety
Use of Resources	Larger (storage, bandwidth, processing)	Smaller (storage, bandwidth, processing)
Adoption	Historically widespread, still common	Growing rapidly, especially in modern applications
Best Suited For	Wide range of applications, legacy systems	Mobile devices, IoT, modern and secure protocols

Fundamentals of Cryptography

- ✓ The algorithms mentioned above will pass into prehistory with quantum computers. Much of the current data is being collected to be decoded later. Since 2024, NIST has introduced resistant algorithms:
 - ML-KEM (Kyber): Intended for general encryption and key exchange on secure websites. Many browsers are already prepared for this algorithm and can be tested in: [Post-Quantum Key Agreement at Cloudflare](#)
 - ML-DSA (Dilithium): Used for digital signatures, ensuring the authenticity of documents and communications.
 - SLH-DSA (SPHINCS+): A digital signature alternative based on hash functions, useful for redundancy and long-term security.

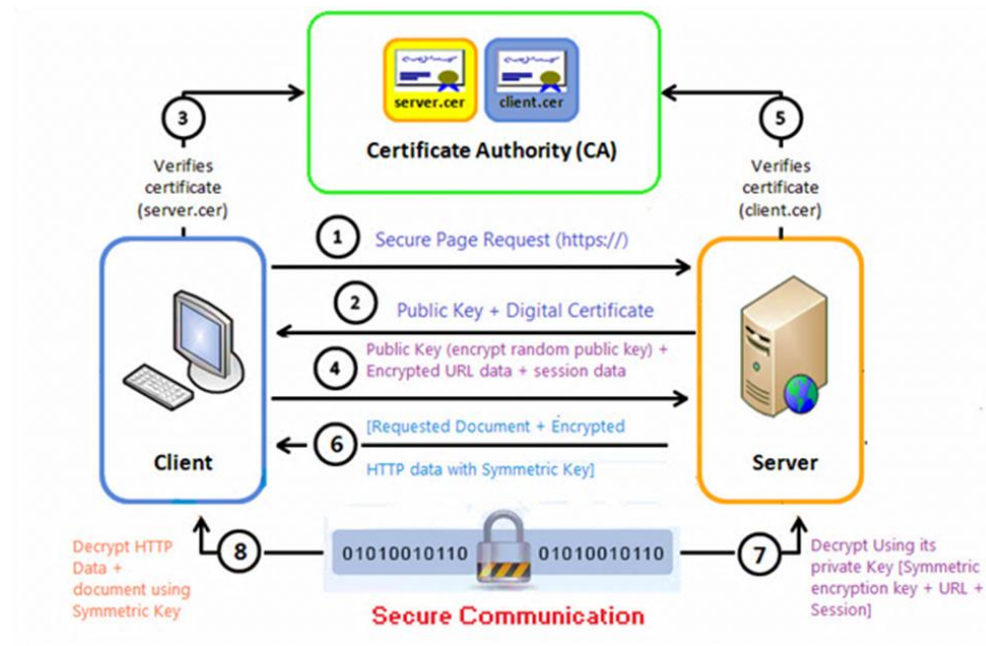
Fundamentals of Cryptography

✓ Examples:

- Digital signature. Whoever signs is the holder of the private key (e.g., is in the CC) whoever verifies the signature has access to the public key
- Handshake of a secure HTTPS (TLS) website. The one who encrypts is the holder of the public key. The public key is sent by the website.

```
-----BEGIN RSA PRIVATE KEY-----
MIIJjwIBAAKCAgEAXYA09A/v6wxgLgp/ULWABPqKA6Eu02pPds1Z0FBvwQpuJMHb
17DjA6HwYheCuMsb+29WB1KuE5OuyKwJhNYrbHkIayT1UktOxdruZmgK8JLknhIn
5U+/p1wB8FFnRuYza1cTrfVq5zFHe/NPsOUyYIc9RupyqsVzWjw/Zg1XhGX5V9HJ
zM8HBz0wHoOrkda700EgsZS91Jhm2P5PYVGWl+ADrbH1VYdtw9LJWmlZXt/gBGgY
1zXsRL1iVhQjvw3Pkm8cAY8HhUrP5s1s72YqNywhH8CGJ+5sJ6aQ7pJLC9HTNmCN
t0z4ADp1PJ2dcMHAmOSUNoIS/KvFwBIuFwJ8nT9Kr0mXSjwyUyN4jI+dw6Ie0MOM
jAFs8+fCMay8UH1119AqngYeZ8TKytjoF+agpjWu0yJhv5JUHvP3gk/yRpUn/XKJ
nJzaCx8VTguRHBQgxVWandhNVpOwm3TLcBJvqa4ZiB1MKpCZryjK7CTF1uCiK5VW
JrabtpD8EuRMat6yC510pM7rCCbIzAKhNQzUNV1b3I3ZdNlotkFP1eHE4lr1tSRc
wJcS0r8Zk70cQ2XZEtVJAgC8lDa10LwvGVEMvfPXOTFHIKB7iyrUC6a8zqv/9KFS
YR+AuqVDBsTbCeYnecPjsiJ4yf7XK5QbLGGn8v/qsTFuqIweHTm8FFFSR0CAwEA
AQKCAgBrCUap1xGRs41Upn/Ff/g3uoid2vcDwTP7zrgMO+u1FTAIX7/hAA6oY1Vh
9Bf2iq1JoVdf6eNj3Nn3CD3UXQX/rcCwQMFrrkCs3jAh8/U2JJqULCqiIP38ddj
JNryA8S0Qgz6VG1xirWRKZZg6mh6FUFSc7Fdd9FCbKqkZB+4M4Pk41+ZzgGjHLg9
cWLEkTo0Ddf0JhJwu//RT1I7J/4zpUaZFjaSGemVHMHezAWRc4d3G+uKh0DbKEBp
```

Fundamentals of Cryptography



<https://resources.infosecinstitute.com/topics/general-security/ssl-dot-net-volume-1-hypothesis/>

- There is a need to verify that the "site is the owner" of the public key. This verification is carried out by:

Fundamentals of Cryptography

- Certificate: Verifies that an entity owns the public key it presents. That is, the certificate contains: the public key, in which situations the public key can be used, who certifies that it is trusted (CA – *Certification authority*) this certification is carried out by signing the CA (using the private key). If the certificate is not signed by a CA, which the browser recognizes, a certificate error is issued (with an error message that scares anyone).



Informações do Certificado

Este certificado é destinado para os seguintes fins:

- Comprova a sua identidade a um computador remoto
- Garante a identidade de um computador remoto
- 2.23.140.1.2.1

Emitido para: *.oliveirasdocarvalho.net

Emitido por: R3

Válido de 13/02/2024 até 13/05/2024

- ✓ Login to Secure shell (SSH) more specifically with openssh: The user has the private key in his possession, the server the public key.
 - Public key is stored in the `~/.ssh/authorized_keys` directory. on the SSH server computer

Fundamentals of Cryptography

- The private key is stored on the SSH client's computer and optionally protected by a password. This additional protection allows you to implement a multi-factor *authentication* process

symmetric and asymmetric encryption

✓ TLS (*Transport Layer Security*)

- The most widely used protocol in communications security. The predecessor was called SSL, which leads to confusion of acronyms.
- Communication begins with a *handshake* based on asymmetric cryptography.
- During the *handshake*, a private key is created. This key allows for symmetric encryption for the rest of the session.

Fundamentals of Cryptography

✓ Wireguard (VPN)

- Communication begins with a *handshake* between the client and the server, called *instances* or *nodes*) based on asymmetric cryptography.
- During the *handshake*, a private key is created. This key allows for symmetric encryption between *nodes*. The *server node* has both the public and private key. The *client node* holds the public key. That is, unlike TLS, the public key is not sent.

The image shows a configuration interface for a Wireguard instance. It contains the following fields and values:

- Name:** cibiot-instance
- Instance:** 0
- Public key:** /W4z/wECqkolDsb4rG7f0lWnITibUSamzKl7lFBqeHE=
- Private key:** kIR+MozcGJdFpQ9Za916YQ1eHxu8Y8lwtDwNr1b73A=
- Listen port:** (empty)
- Tunnel address:** 10.10.10.1/32
- Depend on (CARP):** None
- Peers:** cibiot-peer

Below the Tunnel address field, there are links for 'Clear All', 'Copy', and 'Paste'. Below the Peers field, there is a 'Clear All' link.

- Each inter-node link can support n cross-device links (*interfaces* or *peers*).

Fundamentals of Cryptography

- Peer-to-peer calls are carried out in asymmetric cryptography. The *server node* holds the public key of the peer clients. The *client node* holds the public and private key of the *client* peer

Name	cibiot-peer
Public key	1whxVRVSlm6hR49yLdOYUyX7PtAb935UImf5zKm2l...
Pre-shared key	
Allowed IPs	10.10.10.16/32   Clear All  Copy  Paste
Endpoint address	
Endpoint port	
Instances	cibiot-instance   Clear All

Hash Functions

- ✓ A HASH function is not an encryption, it is a mathematical function that applied to a text returns a fixed-length sentence that "represents" the text
- ✓ A change in the text necessarily implies a change in the hash phrase
- ✓ The hash function is a basic tool in cryptography because it ensures the veracity of the text
- ✓ It is extremely difficult to reverse engineer a hash function
- ✓ Most Used Hash Calculation Algorithms MD5, SHA-1, SHA-256

Fundamentals of Cryptography

Hash Function	Output (Hash Value)
MD5 (128-bit, 16-byte) 32 characters	3A10 0B15 B943 0B17 11F2 E38F 0593 9A9A
SHA-1 (160-bit, 20-byte) 40 characters	569D C9F0 7B48 7F58 9241 AD4C 5C28 7DA0 A448 8D08
SHA-256 (256-bit, 32-byte) 64 characters	F3ED 0867 48FF 3641 3091 0BB6 6293 7080 2958 B5A2 52AF F364 1FC5 07FD E80D 9929

<https://corporatefinanceinstitute.com/resources/cryptocurrency/hash-function/>

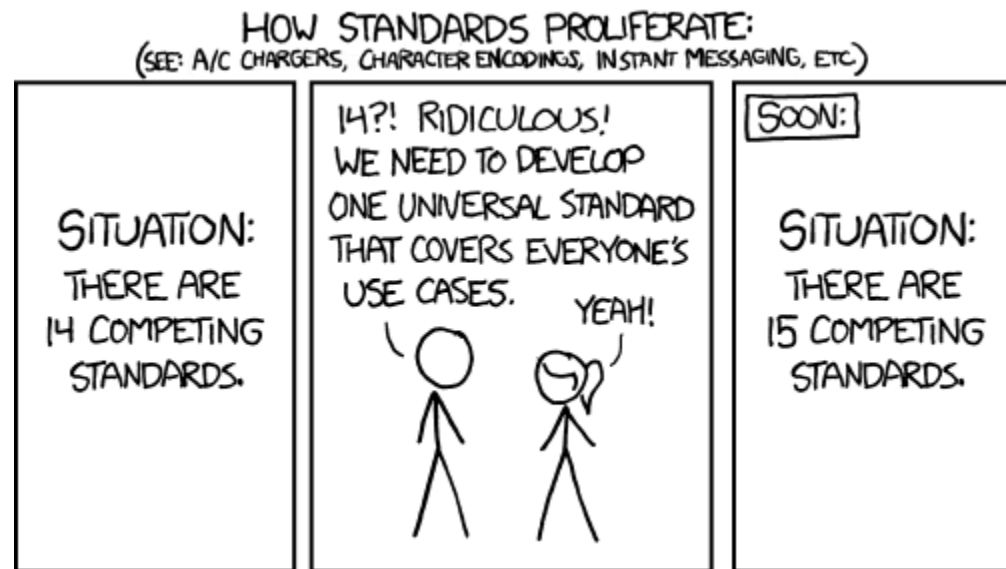
✓ Examples

- Bitcoin – Uses sha-256 (mining is nothing more than finding the number that added to a block returns a *hash* with a certain number)
- Software authenticity check
- E-Invoice. (in this case the *hash* is calculated with reference to all previous invoices)
- Diplomas do ISEP

Fundamentals of Cryptography

Applying Encryption in IOT

- ✓ In 2016, the *mirai botnet* infected thousands of IoT devices by using the fact that communications were not encrypted and authentications were factory.
- ✓ MQTT (Layer 7 Enforcement)
 - Use of authentication using asymmetric keys.
 - Use TLS encryption on the communication channel
- ✓ BLE (*Bluetooth low energy*)
 - Ensure the use of version 4.2 that uses symmetric encryption with a long key after a *handshake* in asymmetric encryption.



Fundamentals of Cryptography

✓ Thread (MATTER)

- Wi-Fi-based, mesh configuration, using a unique key across the network for symmetric encryption.
- The key is distributed to a new element of the network *thread* during a *handshake* in asymmetric encryption
- Before connecting a new element to the network, the identification of this new element must be integrated into the network.

✓ zigbee, zwave, xmpp, lwm2m, lora, lora-wan, dds, coap, gsm, amqp, fm airwaves

- See image above

✓ Does everyone use encryption? They make it possible, now if it is used...